

Question Bank – Module 4: Developing IoT

No.	Question	Marks	Bloom's Taxonomy Level
1	Define IoT Platform. Explain its role in IoT architecture.	5	Remembering
2	List and explain the components of an IoT Platform.	5	Remembering
3	What are Python packages of interest for IoT development? Give examples.	5	Remembering
4	Describe JSON and its importance in IoT data exchange.	6	Understanding
5	List any four advantages and disadvantages of IoT.	5	Remembering
6	Explain the use of RPi.GPIO in IoT applications.	5	Applying
7	Mention the significance of the Edge Layer in IoT architecture.	5	Understanding
8	Describe the ten-step IoT system design methodology.	6	Understanding
9	Explain the concept of "Domain Model Specification" in IoT Design Methodology.	6	Analyzing
10	Describe "Information Model Specification" and its relevance in IoT Design.	6	Analyzing
11	Discuss the "Service Specification" stage and its importance in IoT design.	6	Analyzing
12	Illustrate IoT Functional View and Operational View with suitable examples.	6	Understanding
13	Explain IoT Level Specification and its impact on scalability.	6	Analyzing
14	Explain the architecture of an IoT-based Weather	6	Applying



	Monitoring System.		
15	Describe the role of Python's data processing libraries (NumPy, Pandas) in IoT.	6	Applying
16	Compare JSON and XML for IoT data handling.	6	Analyzing
17	Discuss the importance of cloud platforms in IoT data management.	6	Understanding
18	Describe the Cloud Platform Layer and its main components in IoT.	6	Understanding
19	Explain the middleware layer and its role in IoT platforms.	6	Evaluating
20	Discuss Python's suitability for IoT as an interpreted, interactive, and portable language.	6	Evaluating
21	Analyze the role of each layer in an IoT Platform stack with examples.	8	Analyzing
22	Differentiate between Edge Computing and Cloud Computing in IoT.	8	Analyzing
23	Explain with examples how IoT devices and components are integrated.	8	Applying
24	Compare IoT communication protocols: Wi-Fi, LoRa, ZigBee, and 5G.	8	Analyzing
25	Explain the case study of a Smart Agriculture IoT System.	8	Applying
26	Write and explain Python file handling operations used in IoT applications.	8	Applying
27	Evaluate the significance of Middleware Layer in IoT Platforms.	8	Evaluating
28	Describe the integration of AWS IoT Core and AWS Lambda in Weather Monitoring.	8	Applying
29	Discuss the importance of IoT security and suggest preventive	8	Evaluating



30	Design and explain IoT-based system architecture for Smart Home Application.	10	Creating
31	Develop an IoT Design Methodology for an Industrial Monitoring System.	10	Creating
32	Evaluate IoT Design Methodology in terms of scalability and reliability.	10	Evaluating
33	Explain the integration of Machine Learning in IoT analytics and prediction.	10	Creating
34	Develop a complete IoT Weather Monitoring dashboard architecture covering all layers.	10	Creating
35	Formulate a real-time IoT use case scenario using Python, explaining data flow and processing.	10	Creating